

Branch Specialization in Attention

Toy Induction Head-Dimension Intervention Checkpoint

Research checkpoint

May 22, 2026

RQ5: Does structural heterogeneity cause more stable functional specialization?

This checkpoint tests whether the key-value toy result generalizes to a repeated-token induction-style task:

$$[x_1, \dots, x_{16}, x_1, \dots, x_{16}]$$

Loss is scored on second-half next-token prediction.

Experimental Arms

All models use $d_{model} = 128$, two layers, five seeds, and total attention head dimension 128.

Config	Head dimensions	Purpose
uniform4	[32,32,32,32]	same-head-count baseline
hetero4	[16,16,32,64]	heterogeneous intervention
uniform2	[64,64]	fewer/wider control
64-first	[64,16,16,32]	position control

Primary specialization score:

$$S(h, t) = \max(\text{loss}_{\text{ablate}(h)} - \text{loss}_{\text{base}}, 0)$$

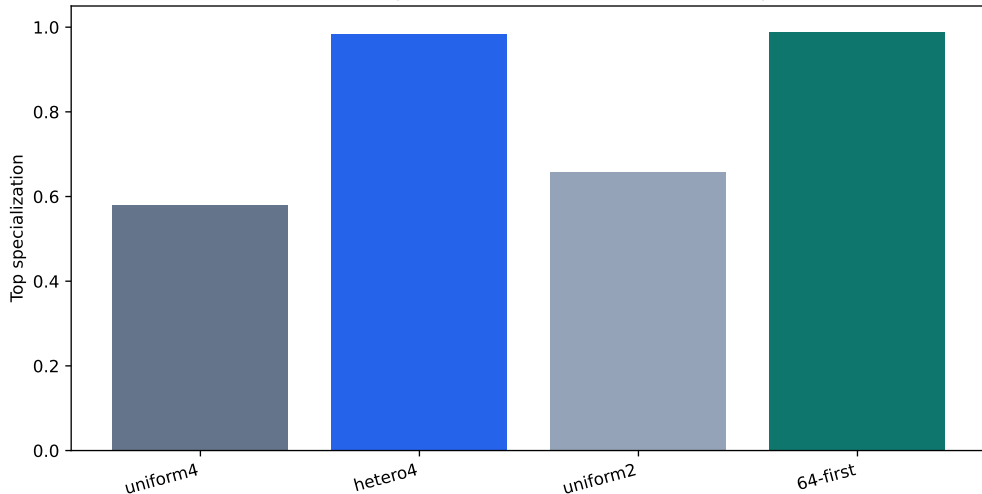
The script also records match and source-next attention, but the main specialization metric is causal single-head ablation.

Config	Eval accuracy	Eval loss
uniform4	0.9993	0.0044
hetero4	0.9991	0.0056
uniform2	0.9993	0.0044
64-first	0.9992	0.0053

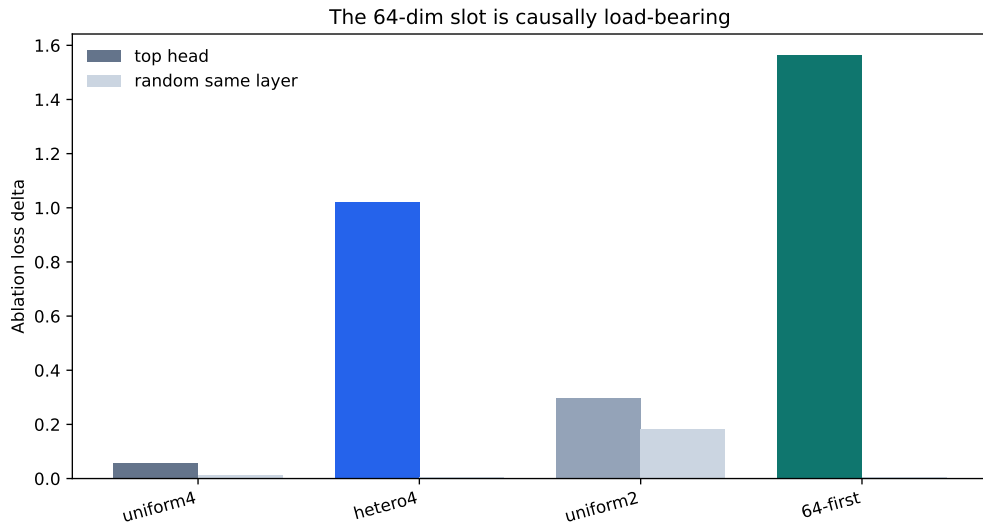
The comparison is not driven by task failure.

Specialization Concentration

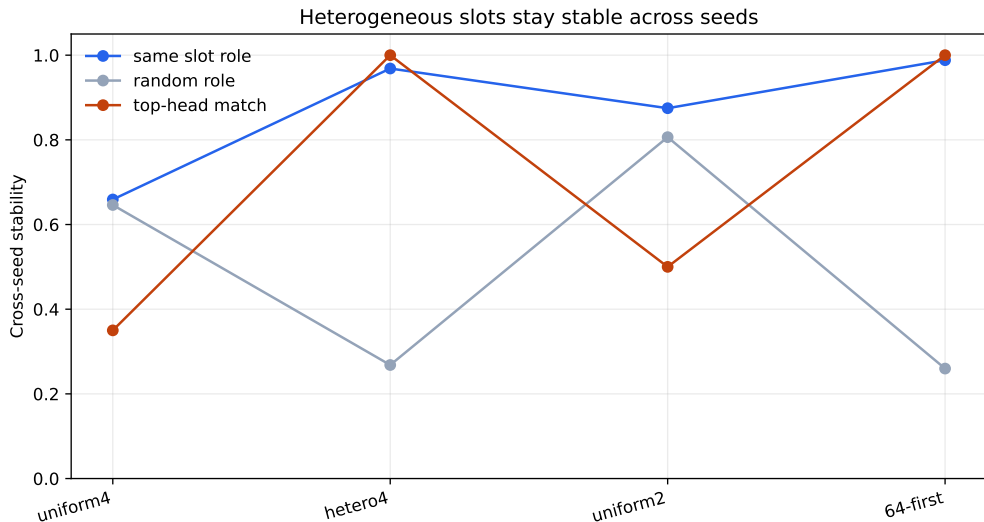
Induction task: heterogeneous heads concentrate causal specialization



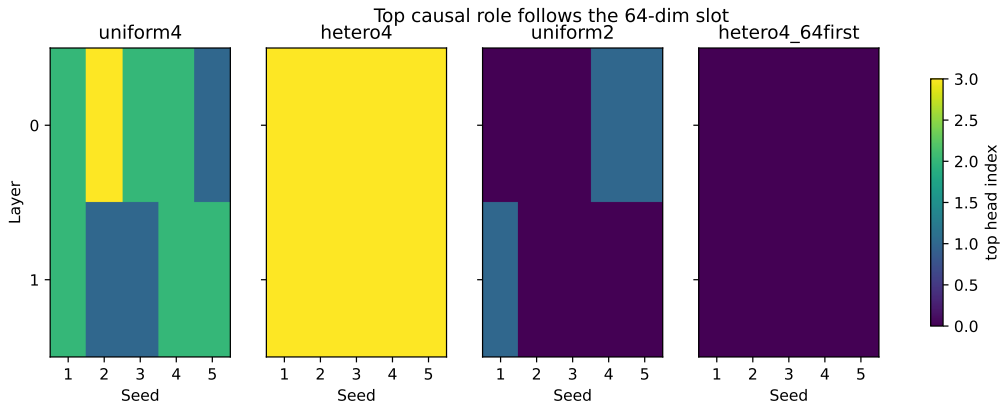
Causal Load



Cross-Seed Stability



Position Control



Findings:

- The heterogeneous result generalizes from key-value recall to induction-style next-token prediction.
- The 64-dim head is the top causal head in both layers and all five seeds for both heterogeneous layouts.
- Uniform wider/fewer heads help, but do not define the same stable structural slot.

Continue the architectural-intervention branch.

Next decisive experiment:

- add task competition: local previous-token plus global induction;
- test whether smaller heads take local roles and the 64-dim head takes global roles;
- report structure-aware slot stability next to Hungarian alignment.

- Script: `scripts/toy_induction_head_dim_intervention.py`
- Output: `results/phase3_toy_induction_head_dim_intervention`
- Report: `doc/phase3_toy_induction_head_dim_intervention.md`
- Data copied into this deck folder under `data/`